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§4. Solvable groups, simplicity of A_n ($n \geq 5$), finitely generated abelian groups

Solvable groups.

DEF. A group G is called solvable if there is a chain of subgroups

$$\{e\} = G_0 \triangleleft G_1 \triangleleft \cdots \triangleleft G_k = G, \text{ st. } G_i/G_{i-1} \text{ is abelian, } \forall 1 \leq i \leq k.$$

COR. 1) For G a finite group,

Jordan-Hölder

交换群只能是 $\mathbb{Z}/p\mathbb{Z}$

→ 证: 设 G 为交换群, 则 G 的子群为正规.

G is solvable $\Leftrightarrow$ all composition factors of G are of the form $\mathbb{Z}/p\mathbb{Z}$, p prime.

任取 $e \neq g \in G$, 则 $\langle g \rangle \triangleleft G$, 则 $G = \langle g \rangle$.

2) Let G be a group and let $N \triangleleft G$.

若 G 无限, 则 $G \cong \mathbb{Z}$, 但 $2\mathbb{Z} \triangleleft \mathbb{Z}$, 不单.

If N and G/N are both solvable, then G is also solvable. (lattice isom. thm.)

若 $\#G$ 为合数, 则必有非平凡真子群. 故 $G \cong \mathbb{Z}/p\mathbb{Z}$.

eg. abelian groups are solvable.

nonabelian simple groups are not solvable.

(in particular, A_5 is not solvable.)

$$G = \left\{ \begin{pmatrix} * & * & * \\ 0 & * & * \\ 0 & 0 & * \end{pmatrix} \in GL_3(\mathbb{C}) \right\} \text{ upper triangular invertible matrices}$$

$$\downarrow \quad N = \left\{ \begin{pmatrix} 1 & * & * \\ 0 & 1 & * \\ 0 & 0 & 1 \end{pmatrix} \right\} \quad G/N \cong (\mathbb{C}^\times)^3$$

$$\downarrow \quad N' = \left\{ \begin{pmatrix} 1 & 0 & * \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right\} \cong \mathbb{C} \quad N/N' \cong \mathbb{C}^2$$

Alternating groups. A_n . $\{e\} \triangleleft A_n \triangleleft S_n$ composition series for $n \geq 5$.

Recall. Every single $(a_1 a_2 \cdots a_n) \in S_n$ is a product of transpositions

$$(a_1 \cdots a_n) = (a_1 a_n)(a_1 a_{n-1}) \cdots (a_1 a_2)$$

$$\text{Consider } \Delta := \prod_{1 \leq i < j \leq n} (x_i - x_j). \quad \sigma(\Delta) := \prod_{1 \leq i < j \leq n} (x_{\sigma(i)} - x_{\sigma(j)}) = \pm \Delta.$$

$$\sigma(\Delta) = \text{sgn}(\sigma) \Delta. \quad \sigma \text{ is } \begin{cases} \text{even, if } \text{sgn}(\sigma) = 1 \\ \text{odd, if } \text{sgn}(\sigma) = -1. \end{cases}$$

LEM. $\text{sgn}: S_n \rightarrow \{\pm 1\} \cong \mathbb{Z}/2\mathbb{Z}$ is a homomorphism.

DEF. $A_n := \ker(\text{sgn}: S_n \rightarrow \{\pm 1\})$ is called the alternating group.

交错群

$$A_n \triangleleft S_n. \quad S_n/A_n \cong \{\pm 1\} \quad \#A_n = \frac{\#S_n}{2} = \frac{n!}{2}.$$

THM A_n is a nonabelian simple group for $n \geq 5$.

REM $A_1, A_2 = \{e\}$. $A_3 = \langle (123) \rangle$ cyclic for order 3.

$$A_4 \supset \{1, (12)(34), (13)(24), (14)(23)\}.$$

Any simple group of order 60 is isomorphic to A_5 (will probably see).

Pf. Call $(ijk) \in A_n$ a 3-cycle.

$$\forall \sigma \in S_n, \sigma(a_1 a_2 \dots a_m) \sigma^{-1} = (\sigma(a_1) \sigma(a_2) \dots \sigma(a_m)) \quad (\sigma(a_i) \mapsto a_i \mapsto a_{i+1} \mapsto \sigma(a_{i+1}))$$

Step 1. A_n is generated by 3-cycles.

↓
elements of the form $(ab)(cd)$ and $(ab)(ac)$

$$\text{But } (ab)(cd) = (acb)(acd) \quad (ab)(ac) = (acb)$$

Step 2. If $N \triangleleft A_n$ contains one 3-cycle (ijk) then it contains all 3-cycles.

$$\text{Want } \forall \sigma \in S_n, (\sigma(i)\sigma(j)\sigma(k)) \in N.$$

$$\text{If } \sigma \in A_n, \text{ then } \sigma(ijk)\sigma^{-1} = (\sigma(i)\sigma(j)\sigma(k)) \in N$$

$$\text{if } \sigma \notin A_n, \text{ then } \sigma(ij) \in A_n \Rightarrow \sigma(ij)(ijk)(ij)\sigma^{-1} = \sigma(jik)\sigma^{-1} = (\sigma(j)\sigma(i)\sigma(k)) \in N.$$

$$\Rightarrow (\sigma(i)\sigma(j)\sigma(k)) = (\sigma(j)\sigma(i)\sigma(k))^2 \in N.$$

Step 3. If $\{e\} \neq N \triangleleft A_n$ ($n \geq 5$), then N contains a 3-cycle.

Fix $e \neq \sigma \in N$.

1) if σ is a product of disjoint cycles

and at least one cycle has length ≥ 4 .

$$\sigma = \mu(a_1 a_2 \dots a_r) \quad r \geq 4 \Rightarrow (a_1 a_2 a_3) \sigma (a_1 a_2 a_3)^{-1} = \mu(a_2 a_3 a_1 a_4 \dots a_r) \in N$$

$$\text{Compute } \sigma^{-1}((a_1 a_2 a_3) \sigma (a_1 a_2 a_3)^{-1}) = (a_1 a_2 \dots a_r)^{-1} \cancel{\mu} \cancel{\mu} (a_2 a_3 a_1 a_4 \dots a_r) = (a_1 a_3 a_r) \in N.$$

2) else $\Rightarrow \sigma$ is product of disjoint cycles of length of 2 and 3.

$\Rightarrow \sigma^3$ is a product of disjoint transpositions

σ^2 ————— 3-cycles

and σ^2, σ^3 can not be both trivial.

→ 否则 $e = \sigma^3(\sigma^2)^{-1} = \sigma$, 矛盾.

replace σ by σ^2 or σ^3 .

3) if σ is a product of (at least 2) disjoint transpositions

$$\sigma = \mu(a_1 a_2)(a_3 a_4) \Rightarrow (a_1 a_2 a_3) \sigma (a_1 a_2 a_3)^{-1} \sigma^{-1} = (a_1 a_3)(a_2 a_4) =: \tau \in N.$$

$n \geq 5$ 用在此处

Then use $n \geq 5$ to take $a_5 \in \{1, 2, \dots, n\}$ different from $a_1, \dots, a_4$

$$\Rightarrow (a_1 a_2 a_5) \tau (a_1 a_2 a_5)^{-1} \tau^{-1} = (a_1 a_2 a_5 a_4 a_3) \in N. \text{ Back to 1).}$$

4) if σ is a product of (at least 2) disjoint 3-cycles

$$\sigma = \mu(a_1 a_2 a_3)(a_4 a_5 a_6) \Rightarrow (a_1 a_2 a_4) \sigma (a_1 a_2 a_4)^{-1} \sigma^{-1} = (a_1 a_2 a_5 a_3 a_4) \in N. \text{ Back to 1). } \square$$

似乎反复出现 $\alpha \beta \alpha^{-1} \beta^{-1}$. 有线索吗?

$\beta \in N, \alpha \in A_n$ 正规性

$$\alpha \beta \alpha^{-1} \in N \Rightarrow \alpha \beta \alpha^{-1} \beta \in N.$$

$\alpha \beta \alpha^{-1} \beta^{-1}$ 为换位子. 作用:

$\alpha \beta \alpha^{-1}$ 是给 β 重新编号, 若 α 只动少数几个点,

则与原来的 β 有很多相同部分, 乘上 β^{-1} 就能消去无关部分,

只剩一个简单的置换.

Finitely generated abelian groups

Recall. A group G is finitely-generated if $\exists$ finite subset A of G st. $G = \langle A \rangle$.

THM (Classification of f.g. abelian groups). Let G be a f.g. abelian groups.

Then $G \cong \mathbb{Z}^r \times \mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z} \times \dots \times \mathbb{Z}/n_k\mathbb{Z}$ for integers $r \geq 0, 2 \leq n_1 | n_2 | \dots | n_k$.

r free part
called the (free) rank of G .

Moreover, such $r, n_1, \dots, n_k$ are unique.

divide

